Sheng Wan

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Education

Shanghai Jiao Tong University, Computer Science — Shanghai, China
- Master's degree in Computer Science (dual degree), expected 2028 — 2026 – 2028

École Polytechnique, Applied Mathematics & Computer Science — Paris, France
- Diplôme d'Ingénieur (Master's level), expected 2027 — 2024 – 2027

Shanghai Jiao Tong University, Information Engineering; French Literature — Shanghai, China
- B.Eng. in Information Engineering — 2022 – 2026
- B.A. in French Literature

Publications

Benchmarking Machine Learning for Chaotic Three-Body Dynamics in Astrophysics — 2026
Sheng Wan, Pavan Vynatheya, Duo Xu, Kevin Li, Ye Zhu
openreview.net/forum?id=qv1Vw0oU7O (NeurIPS 2026 Workshop: Sim2Science: ML with Imperfect Scientific Models)

Experience

Shanghai Luoshu Investment, Quantitative Research Intern — Shanghai, China
June 2026 – Aug 2026
4 months
- Volatility modeling & systematic options trading: built an end-to-end minute-level volatility research pipeline for commodity futures options, including IV calibration, Greeks estimation and volatility surface construction; developed and backtested delta-hedged straddle strategies driven by IV–RV signals.
- Treasury cash-futures relative value: developed a framework mapping Chinese government bond futures to cash bond fair values through CTD, conversion factor, basis and implied-repo analytics; constructed duration/DV01-neutral relative-value signals across TS/TF/T/TL and cash bonds.

Inria, Research Intern — Saclay, France
July 2025 – Aug 2025
3 months
- ANaGRAM for PINNs optimization: applied ANaGRAM, an adaptive natural-gradient method based on the Neural Tangent Kernel (NTK), to improve the convergence speed and accuracy of Physics-Informed Neural Networks (PINNs).
- Adaptive collocation point selection: developed a collocation-embedding-guided adaptive sampling strategy for ANaGRAM, achieving faster convergence and higher accuracy during NTK-based projection.

A.O. Smith, Software Developer Intern — Nanjing, China
July 2023 – Aug 2023
3 months
- IoT data platform: built a real-time MQTT-based IoT data pipeline using Java Spring, with configurable processing through reflection and concurrency-safe operations via Redis distributed locks.
- Database optimization: resolved SQL pagination issues and optimized queries, improving data retrieval reliability and performance.

Awards

China Collegiate Programming Contest (CCPC), Shanghai Region — Silver Prize — 2022
CCPC

National Olympiad in Mathematics in Provinces, Jiangsu — Second Prize — 2021
Jiangsu Province

National Olympiad in Physics in Provinces, Jiangsu — Second Prize 2021
Jiangsu Province

National Olympiad in Informatics in Provinces (NOIP), Jiangsu — First Prize 2020
Jiangsu Province

Skills

Programming

ML & Data

Tools

Languages

Chinese
Native

English
C1

French
C1